

Problem

Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1 .

Input Format

The first line contains an integer T , the number of test cases. For each test case: The first line contains two integers V and N . V — the target value. N — the number of available coin denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

 $1 \leq T \leq 100$ $1 \leq V \leq 10^6$ $1 \leq N \leq 10^6$ $1 \leq C[i] \leq 10^6$

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1 .

Sample Input 0

Change Theme Language Python 3

```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 t,n=map(int,input().split())
3 c=list(map(int,input().split()))
4 dp=[float('inf')]*(t+1)
5 dp[0]=0
6 for i in range(1,n+1):
7     for j in range(1,t+1):
8         dp[j]=min(dp[j],dp[j-c[i-1]]+1)
9 print(dp[-1] if dp[-1]!=float('inf') else -1)
10
```

Line: 10 Col: 1



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Test against custom input

Discussions

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